Projects Report for Molecular Modeling in Process Engineering

Author:
Giorgio Venezia
Student ID: 10807335

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This report presents a comprehensive portfolio of the projects carried out during the course of Molecular Modeling in Process Engineering.

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1 Project 1: Estimation of quantized properties

This project applies fundamental quantum mechanics to molecular and atomic systems. It aims to determine the translational zero-point energy of a methane molecule in a box, calculate hydrogen atomic orbital energies both theoretically and via quantum chemistry, plot orbitals on the xy plane, and find the average electron–nucleus distance for different hydrogen orbitals.

1.1 Determine the translational zero-point energy of a methane molecule in a box having the dimension of 1 m.

To determine the translational zero-point energy of a methane molecule confined within a three-dimensional cubic box of dimension $L = 1\,\text{m}$, we begin with the time-independent Schrödinger Equation:

$$\hat{H}\Psi(\vec{r}) = E\Psi(\vec{r}), \tag{1}$$

where $\vec{r} = (x, y, z)$ represents the Cartesian coordinates of the center of mass of the methane molecule, and m is its total mass. The Hamiltonian operator $\hat{H}$ is defined as the sum of the kinetic energy operator $\hat{T}$ and the potential energy operator $\hat{V}$:

$$\hat{H} = \hat{T} + \hat{V} = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) + V(\vec{r}). \tag{2}$$

The potential energy function $V(\vec{r})$ models the impenetrable walls of the cubic box and is defined as:

$$V(x, y, z) = \begin{cases} 0 & \text{if } 0 < x < L,\ 0 < y < L,\ 0 < z < L \\ \infty & \text{otherwise.} \end{cases} \tag{3}$$

Outside the boundaries of the box, where $V(\vec{r}) = \infty$, the probability of finding the particle must be zero, forcing the wavefunction to vanish ($\Psi(\vec{r}) = 0$). Inside the box, where $V(\vec{r}) = 0$, the Schrödinger equation simplifies to:

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\Psi(\vec{r})}{\partial x^2} + \frac{\partial^2\Psi(\vec{r})}{\partial y^2} + \frac{\partial^2\Psi(\vec{r})}{\partial z^2}\right) = E_{\text{tr}}\Psi(\vec{r}), \tag{4}$$

where E_{tr} is the total translational energy.

By applying the method of separation of variables, the total wavefunction can be expressed as the product of three independent one-dimensional functions, $\Psi(x, y, z) = \psi_x(x)\psi_y(y)\psi_z(z)$, allowing the total energy to be split into its spatial contributions ($E_{\text{tr}} = E_x + E_y + E_z$). For instance, along the x-axis, the equation reduces to the following ordinary differential equation:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_x}{dx^2} = E_x\psi_x. \tag{5}$$

The general solution to this expression is $\psi_x(x) = A\cos(kx) + B\sin(kx)$, where A, B, and k are constants. To determine these parameters, we impose the boundary conditions at the walls of the box ($x = 0$ and $x = L$):

$$\begin{cases} \psi_x(0) = A = 0 \\ \psi_x(L) = A\cos(kL) + B\sin(kL) = 0. \end{cases} \tag{6}$$

Since $A = 0$, the second boundary condition simplifies to $B\sin(kL) = 0$. To avoid the trivial solution ($\psi_x = 0$ everywhere), the coefficient B must be non-zero, which requires $\sin(kL) = 0$. This condition is satisfied if:

$$kL = n_x\pi \quad \text{where } n_x \in \mathbb{N}^+ \tag{7}$$

Substituting $k = \frac{n_x\pi}{L}$ back into Equation 5 yields the quantized energy levels for the x-direction:

$$E_x = \frac{n_x^2 h^2}{8mL^2}. \tag{8}$$

Following the same mathematical reasoning for the y and z directions and substituting the results back into Equation 4, we obtain the expression for the total translational energy in a 3D cubic box:

$$E_{\text{tr}} = \frac{h^2}{8mL^2}\left(n_x^2 + n_y^2 + n_z^2\right). \tag{9}$$

The Translational Zero-Point Energy (ZPE) corresponds to the ground state, where the quantum numbers assume their minimum physical values ($n_x = n_y = n_z = 1$). Given that the box dimension is $L = 1\,\mathrm{m}$, the formula reduces to:

$$E_{\mathrm{tr}} = \frac{3h^2}{8m}. \tag{10}$$

The total mass of the methane molecule ($m_{\mathrm{CH_4}}$) is computed from its molecular weight and Avogadro's number:

$$m_{\mathrm{CH_4}} = \frac{(12.011 + 4 \times 1.008) \times 10^{-3}\ \mathrm{kg/mol}}{6.022 \times 10^{23}\ \mathrm{mol^{-1}}} \approx 2.664 \times 10^{-26}\ \mathrm{kg}. \tag{11}$$

Substituting this mass and Planck's constant (h) into the ground-state energy expression yields the final value:

$$E_{\mathrm{tr}} = 6.196\,896\,013 \times 10^{-42}\,\mathrm{J} = 1.421\,388\,513\,447 \times 10^{-24}\,E_{\mathrm{h}}. \tag{12}$$

1.2 Determine the energy of the orbitals of atomic hydrogen both from theory and using a quantum chemistry approach.

For this task, the discussion is divided into a rigorous theoretical analysis and a computational quantum chemistry approach.

Theoretical Approach

The hydrogen atom consists of a proton of mass M and an electron of mass m_e. Their electrostatic interaction is modeled via the Coulomb potential energy function:

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0 r}$$

Assuming the heavier proton is fixed at the origin of the coordinate system, the Hamiltonian operator for the single moving electron is expressed as:

$$\hat{H} = -\frac{\hbar^2}{2m_e}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) - \frac{e^2}{4\pi\varepsilon_0 r}, \tag{13}$$

where e is the elementary charge, ε_0 is the vacuum permittivity, and r is the electron-nucleus distance.

Exploiting the spherical symmetry of the potential, a transformation from Cartesian to spherical coordinates (r, θ, ϕ) is performed, yielding the Schrödinger equation in the following form:

$$-\frac{\hbar^2}{2m_e}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\phi^2}\right] - \frac{e^2}{4\pi\varepsilon_0 r}\psi = E\psi. \tag{14}$$

To solve this partial differential equation, the method of separation of variables is applied by assuming a product solution of a radial function $R(r)$ and an angular function $Y(\theta, \phi)$:

$$\psi(r, \theta, \phi) = R(r)Y(\theta, \phi). \tag{15}$$

Substituting this product into the Schrödinger equation and refactoring allows the radial and angular dependencies to be fully isolated:

$$\frac{1}{R(r)}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{2m_e r^2}{\hbar^2}\left(E + \frac{e^2}{4\pi\varepsilon_0 r}\right) = -\frac{1}{Y(\theta,\phi)}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\phi^2}\right]. \tag{16}$$

Since the left-hand side depends strictly on r and the right-hand side depends strictly on θ and ϕ, both sides must equal a constant, denoted as β. This yields a separated system of equations:

$$\begin{cases} \dfrac{d}{dr}\left(r^2\dfrac{dR}{dr}\right) + \dfrac{2m_e r^2}{\hbar^2}\left(E + \dfrac{e^2}{4\pi\varepsilon_0 r}\right)R(r) = \beta R(r) & (17) \\[2ex] \dfrac{1}{\sin\theta}\dfrac{\partial}{\partial\theta}\left(\sin\theta\dfrac{\partial Y}{\partial\theta}\right) + \dfrac{1}{\sin^2\theta}\dfrac{\partial^2 Y}{\partial\phi^2} = -\beta Y(\theta,\phi) & (18) \end{cases}$$

Solving this system yields $\beta = \ell(\ell+1)$, where ℓ is the azimuthal quantum number ($\ell = 0, 1, \ldots, n-1$) governing the shape of the orbital. With the separation constant established, equations (17) and (18) can be solved independently to yield the complete spatial description of the electron.

The Radial Solution Substituting $\beta = \ell(\ell+1)$ into the radial equation (17) and imposing physical bound-state conditions requires the introduction of the principal quantum number $n \in \mathbb{N}^+$. This parameter mathematically defines the quantized energy levels and sets the upper limit for the angular momentum. The resulting normalized radial wavefunction $R_{n,\ell}(r)$ is expressed analytically in terms of the Associated Laguerre polynomials, $L_{n-\ell-1}^{2\ell+1}$:

$$R_{n,\ell}(r) = \sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-\ell-1)!}{2n[(n+\ell)!]}} e^{-r/na_0} \left(\frac{2r}{na_0}\right)^{\ell} L_{n-\ell-1}^{2\ell+1}\left(\frac{2r}{na_0}\right) \tag{19}$$

where a_0 is the Bohr radius.

The Angular Solution The angular equation (18) is solved by further separating its variables into $Y(\theta,\phi) = \Theta(\theta)\Phi(\phi)$. This step introduces a second separation constant, m^2, where boundary conditions restrict m to integer values. This defines the magnetic quantum number:

$$m = -\ell, -\ell+1, \ldots, 0, \ldots, \ell-1, \ell \tag{20}$$

The total angular solutions are the Spherical Harmonics, $Y_{\ell,m}(\theta,\phi)$, which combine complex exponentials for the ϕ-dependency and Associated Legendre polynomials, $P_\ell^m(\cos\theta)$, for the θ-dependency:

$$Y_{\ell,m}(\theta,\phi) = (-1)^{(m+|m|)/2} \sqrt{\frac{(2\ell+1)}{4\pi} \frac{(\ell-|m|)!}{(\ell+|m|)!}} P_\ell^m(\cos\theta) e^{im\phi} \tag{21}$$

The Total Wavefunction By recombining the independent radial and angular components, the total normalized wavefunction—representing a specific hydrogen atomic orbital—is fully defined by the three spatial quantum numbers (n,ℓ,m):

$$\psi_{n,\ell,m}(r,\theta,\phi) = R_{n,\ell}(r) Y_{\ell,m}(\theta,\phi) \tag{22}$$

Since both the radial and angular parts are individually orthonormal, the total wavefunction satisfies the global normalization condition over the entire three-dimensional space:

$$\int_0^\infty \int_0^\pi \int_0^{2\pi} |\psi_{n,\ell,m}(r,\theta,\phi)|^2 r^2 \sin\theta \, dr d\theta d\phi = 1 \tag{23}$$

Substituting Equation 22 back into the spherical Schrödinger equation (14) yields the expression for the energy eigenvalues E:

$$E = -\frac{m_e e^4}{8\varepsilon_0^2 h^2 n^2} \tag{24}$$

By numerical substitution, the final analytical energy eigenvalues for the first two principal energy levels (defined by n) are obtained as:

$$\begin{aligned} n = 1: &\quad E_1 \approx -2.1799 \times 10^{-18}\,\mathrm{J} = -0.5\,E_\mathrm{h} \\ n = 2: &\quad E_2 \approx -5.449\,75 \times 10^{-19}\,\mathrm{J} = -0.125\,E_\mathrm{h} \end{aligned}$$

1.2.1 Quantum Chemistry Approach

To approximate these energy levels numerically, a computational approach was implemented using the `MOLPRO` software package.

Since atomic hydrogen is an open-shell system containing a single unpaired electron, it exhibits a doublet spin state ($2S+1=2$). Therefore, the simulations were carried out utilizing the Unrestricted Hartree-Fock (*UHF*) method. For a one-electron system, the Hartree-Fock framework is formally free of electron correlation errors, meaning that any discrepancy from the exact analytical energy arises exclusively from the incompleteness of the chosen basis set. To minimize this basis set error, the highly flexible, a very big basis set, namely `aug-cc-pVQZ`, was selected.

The computation yields the ground-state energy ($n=1$) of the hydrogen atom as:

$$E_1 = -0.499\,948\,321\,468\,E_\mathrm{h} \tag{25}$$

Evaluating the energy level for the first excited state ($n = 2$) requires a different computational strategy, as the software must be explicitly instructed to set the atom in an excited electronic configuration. In `MOLPRO`, this is achieved using the card `{uhf; wf, 1, 2, 1}`.

Furthermore, because excited-state orbitals are physically more diffuse than the ground state, the basis set was increased to `d-aug-cc-pVQZ`, to ensure the minimum error. This calculation yields:

$$E_2 = -0.124\,810\,255\,914\,E_\mathrm{h} \tag{26}$$

1.2.2 Results and Discussion

The computational approach was is very accurate, with relative errors of 0.02% for the ground state and of 0.15% for the excited state.

The subtle discrepancies observed between the analytical and computed eigenvalues is due to the usage of the basis set. Quantum chemistry software utilize Gaussian-type orbitals with a spatial dependency proportional to $e^{-\alpha r^2}$ to ensure computational efficiency during integral evaluation, instead of Slather-type orbitals which comes from the theory, with a spatial dependency proportional to $e^{-\alpha r}$.

1.3 Draw the $1s$, $2s$, $2p_x$, $2p_y$, and $2p_z$ Orbitals of Atomic Hydrogen on a xy Plane

To visualize the orbitals I decided to follow a different workflow from the one presented during lecture. Instead of using `gnuplot`, I wrote a standalone `C++` code to evaluate the analytical wavefunctions by taking as input the quantum numbers n,ℓ,m. This code was paired with a `Python` post-processing script using `matplotlib` to generate the drawings.

The `C++` core application evaluates the total wavefunction $\psi_{n,\ell,m}(r,\theta,\phi) = R_{n,\ell}(r)Y_{\ell,m}(\theta,\phi)$ by computing the exact analytical solutions via stable recurrence relations:

- The radial part $R_{n,\ell}(r)$ is calculated using the recurrence relation for the Associated Laguerre polynomials $L_k^\alpha(\rho)$, where $k = n - \ell - 1$ and $\alpha = 2\ell + 1$:

  $$(i+1)L_{i+1}^\alpha(\rho) = (2i + 1 + \alpha - \rho)L_i^\alpha(\rho) - (i + \alpha)L_{i-1}^\alpha(\rho)$$

  where the algorithm iteratively builds the polynomial degree up to $i = k - 1$.

- The angular component $Y_{\ell,m}(\theta,\phi)$ evaluates the Associated Legendre polynomials $P_\ell^m(x)$ (where $x = \cos\theta$) combined with the complex electronic phase $e^{im\phi}$. The stable stable evaluation of high-order polynomials is achieved via the standard three-term recurrence relation:

  $$(i - m)P_i^m(x) = x(2i - 1)P_{i-1}^m(x) - (i + m - 1)P_{i-2}^m(x)$$

  which iteratively computes the stable amplitude up to the target orbital angular momentum $i = \ell$.

To project the wavefunctions onto the xy plane, the spatial domain is sampled on a dense Cartesian grid where the polar angle is strictly constrained to $\theta = \pi/2$, and the azimuthal angle is dynamically computed as $\phi = \mathrm{atan2}(y, x)$.

The transition from the complex mathematical solutions to the real physical chemistry representations is governed within the code by applying the appropriate unitary transformations for $m \neq 0$:

$$\psi_{2p_x} = \sqrt{2} \cdot R_{2,1}(r) \cdot \mathrm{Re}\left[Y_{1,1}\left(\frac{\pi}{2}, \phi\right)\right] \tag{27}$$

$$\psi_{2p_y} = \sqrt{2} \cdot R_{2,1}(r) \cdot \mathrm{Im}\left[Y_{1,1}\left(\frac{\pi}{2}, \phi\right)\right] \tag{28}$$

For $m = 0$, the wavefunction naturally yields a real solution corresponding to the $2p_z$ orbital orientation.

The 2D projections, illustrated in Figure 1, accurately reconstruct the expected orbitals. The isotropic circular profiles characteristic of the $1s$ and $2s$ orbitals are perfectly calculated within the xy plane.

Furthermore, the behavior of the $\ell = 1$ angular momentum shell is properly solved. The $2p_x$ and $2p_y$ orbitals display their distinct directional dumbbell contours aligned along their respective Cartesian axes, separated by a planar node passing through the nucleus. Conversely, the $2p_z$ orbital exhibits a

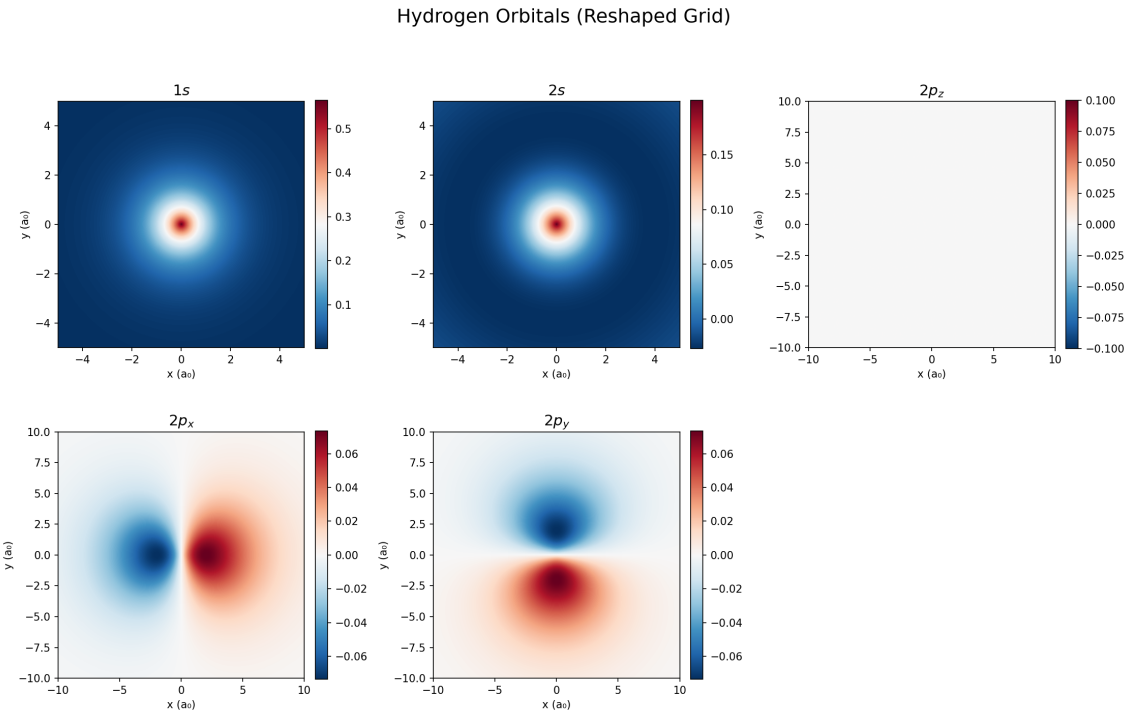


Figure 1: 2D reshaped grid representing the intensity and phase mapping of the $1s$, $2s$, $2p_z$, $2p_x$, and $2p_y$ hydrogen atomic orbitals projected onto the xy plane.

uniformly zero-intensity grid across the entire domain; this is physically consistent with the analytical framework, as the xy projection plane ($\theta = \pi/2$) corresponds exactly to the angular nodal plane of the $m = 0$ wavefunction where $\cos(\pi/2) = 0$.

To enhance the spatial visualization of the atomic states, the `C++` code was extended to map the three-dimensional isosurfaces. By applying a density filtering threshold

$$|\psi|^2 > 0.05 \cdot \max|\psi|^2$$

within the `Python` post-processing framework, the high-probability spatial volumes were successfully isolated and rendered, as shown in Figure 2.

These three-dimensional scatter representations provide a comprehensive overview of the orbital topologies, recovering their characteristic geometric shapes with high fidelity while explicitly capturing the spatial distribution of the underlying quantum phases.

1.4 Determine the average distance of the electron in a 1s, 2s, 2px, 2py, and 2pz orbital from the nuclei for atomic hydrogen.

Hydrogenic wavefunctions are split into radial and angular components using spherical coordinates (r, θ, ϕ):

$$\begin{aligned}\Psi_{nlm}(r,\theta,\phi) &= R_{nl}(r) \cdot Y_l^m(\theta,\phi) \\ &= \left[\left(\frac{2}{na_0}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3} \frac{2l+1}{4\pi} \frac{(l-|m|)!}{(l+|m|)!}\right]^{1/2} \\ &\quad \cdot e^{-\frac{r}{na_0}} \left(\frac{2r}{na_0}\right)^l L_{n+l}^{2l+1}\left(\frac{2r}{na_0}\right) P_l^{|m|}(\cos\theta) e^{im\phi}\end{aligned} \tag{29}$$

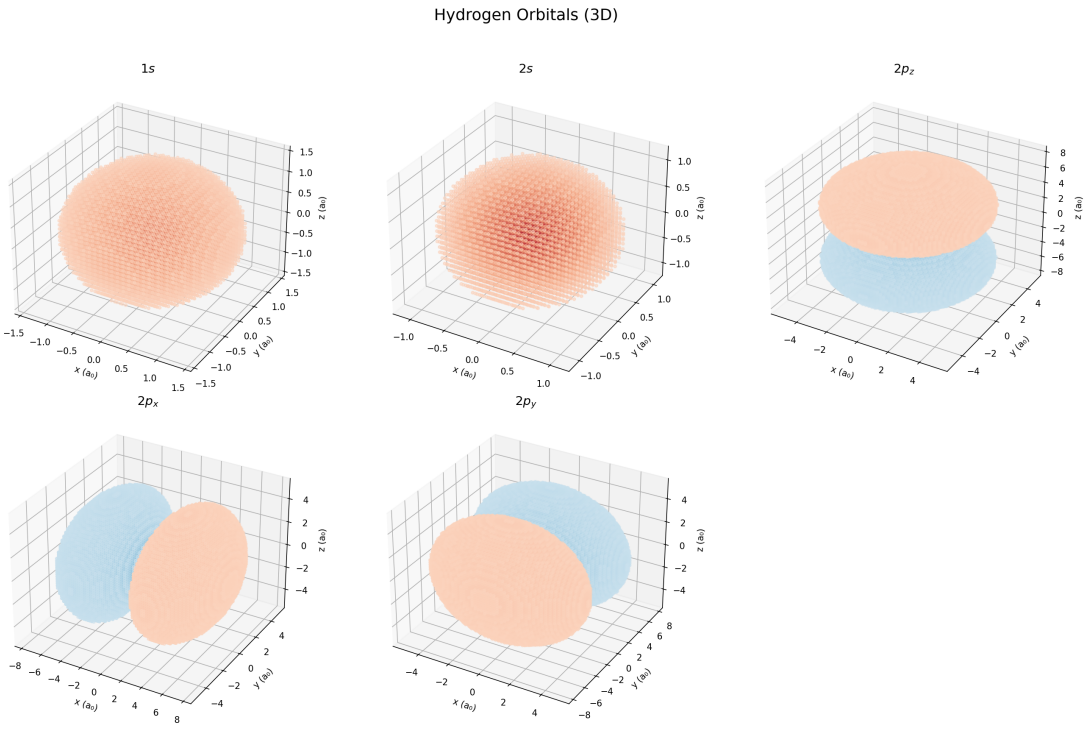


Figure 2: 3D spatial isosurfaces of the evaluated hydrogen atomic orbitals, displaying full spatial geometry and phase configurations (Red $\rightarrow \psi > 0$, Blue $\rightarrow \psi < 0$).

The average distance $\langle r \rangle$ is the expectation value of the position operator $\hat{r}$:

$$\langle r \rangle_{nlm} = \int \Psi^*_{nlm}\, r\, \Psi_{nlm}\, d\tau = \int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta\, d\theta \int_0^{\infty} r^3 |\Psi_{nlm}(r,\theta,\phi)|^2\, dr \tag{30}$$

where the differential volume element is given by $d\tau = r^2 \sin\theta\, dr\, d\theta\, d\phi$.

The fully normalized solutions extracted from the Laguerre and Legendre polynomial expansions are defined below:

$$\Psi_{100} = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0} \tag{31}$$

$$\Psi_{200} = \frac{1}{\sqrt{32\pi}} \left(\frac{1}{a_0}\right)^{\frac{3}{2}} \left(2 - \frac{r}{a_0}\right) e^{-\frac{r}{2a_0}} \tag{32}$$

$$\Psi_{210} = \frac{1}{\sqrt{32\pi}} \left(\frac{1}{a_0}\right)^{\frac{5}{2}} r e^{-\frac{r}{2a_0}} \cos\theta \tag{33}$$

$$\Psi_{21\pm1} = \frac{1}{\sqrt{64\pi}} \left(\frac{1}{a_0}\right)^{\frac{5}{2}} r e^{-\frac{r}{2a_0}} \sin\theta e^{\pm i\phi} \tag{34}$$

The average distance is then computed as:

$$\langle r\rangle_{100} = \int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta\, d\theta \int_0^{\infty} \left(\frac{1}{\pi a_0^3} e^{-\frac{2r}{a_0}}\right) r^3\, dr \tag{35}$$

$$= \frac{1}{\pi a_0^3}\left[\int_0^{2\pi} d\phi\right]\left[\int_0^{\pi} \sin\theta\, d\theta\right]\left[\int_0^{\infty} r^3 e^{-\frac{2r}{a_0}}\, dr\right] \tag{36}$$

$$= \frac{1}{\pi a_0^3}\cdot(2\pi)\cdot(2)\cdot\frac{3!}{\left(\frac{2}{a_0}\right)^4} = \frac{4}{a_0^3}\cdot\frac{6\cdot a_0^4}{16} = \frac{3}{2}a_0 \tag{37}$$

$$\langle r\rangle_{200} = \int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta\, d\theta \int_0^{\infty} r^3 \left[\frac{1}{32\pi a_0^3}\left(2-\frac{r}{a_0}\right)^2 e^{-\frac{r}{a_0}}\right] dr \tag{38}$$

$$= \frac{1}{32\pi a_0^3}\cdot(2\pi)\cdot(2)\cdot\int_0^{\infty} r^3\left(4-\frac{4r}{a_0}+\frac{r^2}{a_0^2}\right)e^{-\frac{r}{a_0}}\, dr \tag{39}$$

$$= \frac{1}{8a_0^3}\left[4\int_0^{\infty} r^3 e^{-\frac{r}{a_0}}\, dr - \frac{4}{a_0}\int_0^{\infty} r^4 e^{-\frac{r}{a_0}}\, dr + \frac{1}{a_0^2}\int_0^{\infty} r^5 e^{-\frac{r}{a_0}}\, dr\right] \tag{40}$$

$$= \frac{1}{8a_0^3}\left[4\left(\frac{3!}{(1/a_0)^4}\right) - \frac{4}{a_0}\left(\frac{4!}{(1/a_0)^5}\right) + \frac{1}{a_0^2}\left(\frac{5!}{(1/a_0)^6}\right)\right] \tag{41}$$

$$= \frac{1}{8a_0^3}\left[24a_0^4 - 96a_0^4 + 120a_0^4\right] = \frac{48a_0^4}{8a_0^3} = 6a_0 \tag{42}$$

$$\langle r\rangle_{210} = \int_0^{2\pi} d\phi \int_0^{\pi} \cos^2\theta \sin\theta\, d\theta \int_0^{\infty} r^3 \left[\frac{1}{32\pi a_0^5} r^2 e^{-\frac{r}{a_0}}\right] dr \tag{43}$$

$$= \frac{1}{32\pi a_0^5}\left[\int_0^{2\pi} d\phi\right]\left[\int_0^{\pi} \cos^2\theta \sin\theta\, d\theta\right]\left[\int_0^{\infty} r^5 e^{-\frac{r}{a_0}}\, dr\right] \tag{44}$$

$$= \frac{1}{32\pi a_0^5}\cdot(2\pi)\cdot\left(\frac{2}{3}\right)\cdot\frac{5!}{\left(\frac{1}{a_0}\right)^6} \tag{45}$$

$$= \frac{4\pi}{96\pi a_0^5}\cdot(120a_0^6) = \frac{480}{96}a_0 = 5a_0 \tag{46}$$

$$\langle r\rangle_{21\pm1} = \int_0^{2\pi} d\phi \int_0^{\pi} \sin^3\theta\, d\theta \int_0^{\infty} r^3 \left[\frac{1}{64\pi a_0^5} r^2 e^{-\frac{r}{a_0}}\right] dr \tag{47}$$

$$= \frac{1}{64\pi a_0^5}\left[\int_0^{2\pi} d\phi\right]\left[\int_0^{\pi} \sin^3\theta\, d\theta\right]\left[\int_0^{\infty} r^5 e^{-\frac{r}{a_0}}\, dr\right] \tag{48}$$

$$= \frac{1}{64\pi a_0^5}\cdot(2\pi)\cdot\left(\frac{4}{3}\right)\cdot\frac{5!}{\left(\frac{1}{a_0}\right)^6} \tag{49}$$

$$= \frac{8\pi}{192\pi a_0^5}\cdot(120a_0^6) = \frac{960}{192}a_0 = 5a_0 \tag{50}$$

2 Project 2: Quantum Chemical Calculation of the structures of complex molecules

2.1 Energy value calculation for C_2H_6

Computational simulations were performed using both Gaussian and Molpro software packages to characterize the structural and energetic properties of ethane (C_2H_6). Density Functional Theory (DFT) calculations were carried out using B3LYP functional combined with the 6-31+G(d,p). Ab initio calculations were alternatively executed at the Hartree-Fock (HF) and MP2 levels using a 6-311G basis set.

The initial structures were constructed using an internal coordinate Z-matrix. Full geometry optimization and subsequent vibrational frequency analyses were performed.

The total electronic ground-state energies (E_{tot}) extracted from the respective convergence outputs are summarized in Table 1:

Table 1: Total Electronic Ground-State Energies (E_{tot}) for Ethane (C_2H_6)

Method	**Basis Set**	**Total Energy (E_{tot} / Hartrees)**
B3LYP	6-31+G(d,p)	−79.84164177
HF	6-311G	−79.21104959
MP2	6-311G	−79.41190613

2.2 Energy changes as function of the relative orientation of the methyl groups

The torsional potential energy surface (PES) profile of ethane was mapped by performing a relaxed scan of the H–C–C–H dihedral angle (d_6) from 300.0° to 360.0° in increments of 10.0°. Table 2 lists the absolute electronic energies calculated at each step, alongside the relative energies (ΔE) reported in both kcal/mol to illustrate the rotational potential barrier.

Table 2: Torsional Energy Profile for Ethane Rotation

Dihedral (d_6/°)	**Total Energy (E_{tot} / Hartree)**	**ΔE (kcal/mol)**
300.0	−79.84164177	0.0000
310.0	−79.84137074	0.1701
320.0	−79.84062504	0.6380
330.0	−79.83956142	1.3054
340.0	−79.83845167	2.0018
350.0	−79.83760766	2.5314
360.0	−79.83728800	2.7320

2.3 Energy value calculation for C_3H_8

Building upon the ethane framework, the molecular system was extended to propane (C_3H_8). The results are in Table 3, full geometric optimizations using these structural variables were successfully processed under B3LYP/6-31+G(d,p) and HF/MP2/6-311G environments to pinpoint the global minimum conformation of propane.

Table 3: Total Electronic Ground-State Energies (E_{tot}) for Propane (C_3H_8)

Method / Functional	**Basis Set**	**Total Energy (E_{tot} / Hartree)**
B3LYP	6-31+G(d,p)	−119.15914567
HF	6-311G	−118.23600305
MP2	6-311G	−118.53505938

3 Project 3: Quantum Chemical based Calculation of partition functions and related quantities

All the computations are performed using the following instruction in gaussian

```
#P B3LYP/6-31+g(d,p) opt=internal freq Pop=Full gfinput
```

Table 4: Calculated Partition Functions and Thermodynamic Properties for CH_4. Note: Constant values across all temperatures omitted for brevity: $Q_{\text{el}} = 1$, $S_{\text{el}} = 0$, $C_{v,\text{trans}} = C_{v,\text{rot}} = 12.47169$.

T (K)	Q_{trans}	Q_{rot}	Q_{vib}	S_{tot}	$C_{v,\text{vib}}$	C_p	$C_{p,\text{NIST}}$	Err (%)
298.15	2.52E+06	36.60	2.55E-21	186.10	2.09	35.35	—	—
300.00	2.56E+06	36.94	3.41E-21	186.32	2.15	35.41	35.76	0.98%
400.00	5.26E+06	56.88	4.58E-16	197.07	6.64	39.90	40.63	1.79%
500.00	9.19E+06	79.49	5.69E-13	206.57	12.26	45.52	46.63	2.38%
600.00	1.45E+07	104.49	6.91E-11	215.38	17.97	51.23	52.74	2.86%
700.00	2.13E+07	131.67	2.24E-09	223.69	23.40	56.66	58.60	3.32%
800.00	2.98E+07	160.87	3.20E-08	231.58	28.43	61.69	64.08	3.74%
900.00	3.99E+07	191.95	2.65E-07	239.12	33.03	66.28	69.14	4.13%
1000.00	5.20E+07	224.82	1.51E-06	246.32	37.19	70.45	73.75	4.48%
1100.00	6.60E+07	259.37	6.50E-06	253.22	40.92	74.18	77.92	4.80%
1200.00	8.20E+07	295.53	2.28E-05	259.82	44.25	77.51	81.68	5.11%
1300.00	1.00E+08	333.23	6.82E-05	266.14	47.20	80.46	85.07	5.42%
1400.00	1.21E+08	372.41	1.80E-04	272.20	49.82	83.07	88.11	5.72%
1500.00	1.43E+08	413.02	4.30E-04	278.01	52.13	85.38	90.86	6.03%
1600.00	1.68E+08	455.00	9.44E-04	283.59	54.17	87.43	93.33	6.33%
1700.00	1.96E+08	498.32	1.94E-03	288.94	55.98	89.23	95.58	6.64%
1800.00	2.26E+08	542.93	3.75E-03	294.09	57.58	90.84	97.63	6.96%
1900.00	2.59E+08	588.80	6.91E-03	299.04	59.00	92.26	99.51	7.29%
2000.00	2.94E+08	635.89	1.22E-02	303.80	60.27	93.53	101.24	7.62%

Table 5: Thermodynamic Properties and Partition Functions for C_2H_6. Note: Constant values across all temperature vectors have been excluded from the core columns for clarity: $Q_{\text{el}} = 1$, $S_{\text{el}} = 0$, $C_{v,\text{trans}} = C_{v,\text{rot}} = 12.47169$.

T (K)	Q_{trans}	Q_{rot}	Q_{vib}	S_{tot}	$C_{v,\text{vib}}$	C_p	$C_{p,\text{NIST}}$	Err (%)
298.15	6.47E+06	812.79	6.52E-35	227.68	17.27	50.52	—	—
300.00	6.57E+06	820.36	1.07E-34	227.99	17.48	50.74	52.71	3.74%
400.00	1.35E+07	1263.03	4.71E-26	244.29	30.17	63.42	65.46	3.11%
500.00	2.36E+07	1765.14	8.68E-21	259.83	43.01	76.27	77.94	2.15%
600.00	3.72E+07	2320.34	3.34E-17	274.80	54.72	87.98	89.19	1.36%
700.00	5.47E+07	2923.96	1.42E-14	289.15	65.11	98.37	99.14	0.77%
800.00	7.63E+07	3572.40	1.53E-12	302.90	74.30	107.55	107.94	0.36%
900.00	1.02E+08	4262.74	6.57E-11	316.05	82.40	115.66	115.71	0.04%
1000.00	1.33E+08	4992.58	1.49E-09	328.61	89.55	122.81	122.55	0.21%
1100.00	1.69E+08	5759.88	2.11E-08	340.62	95.84	129.09	128.55	0.42%
1200.00	2.10E+08	6562.91	2.10E-07	352.09	101.35	134.61	133.80	0.61%
1300.00	2.57E+08	7400.14	1.58E-06	363.06	106.19	139.45	138.39	0.77%
1400.00	3.09E+08	8270.21	9.59E-06	373.55	110.43	143.69	142.40	0.91%
1500.00	3.68E+08	9171.95	4.87E-05	383.60	114.16	147.41	145.90	1.04%
1600.00	4.32E+08	10104.26	2.14E-04	393.22	117.43	150.68	148.90	1.14%
1700.00	5.03E+08	11066.19	8.32E-04	402.44	120.30	153.56	151.67	1.25%
1800.00	5.80E+08	12056.84	2.91E-03	411.29	122.84	156.10	154.04	1.34%
1900.00	6.64E+08	13075.40	9.34E-03	419.79	125.09	158.35	156.14	1.42%
2000.00	7.54E+08	14121.14	2.77E-02	427.97	127.09	160.35	158.00	1.48%

Table 6: Thermodynamic Properties and Partition Functions for H_2O. Note: Constant values across all temperature vectors have been excluded from the core columns for clarity: $Q_{\text{el}} = 1$, $S_{\text{el}} = 0$, $C_{v,\text{trans}} = C_{v,\text{rot}} = 12.47169$.

T (K)	Q_{trans}	Q_{rot}	Q_{vib}	S_{tot}	$C_{v,\text{vib}}$	C_p	$C_{p,\text{NIST}}$	Err (%)
298.15	3.00E+06	43.64	1.61E-10	188.69	0.22	33.48	—	—
300.00	3.05E+06	44.04	1.85E-10	188.90	0.23	33.48	29.04	15.30%
400.00	6.26E+06	67.81	5.04E-08	198.61	0.87	34.13	31.45	8.53%
500.00	1.09E+07	94.77	1.46E-06	206.32	1.82	35.08	33.56	4.52%
600.00	1.73E+07	124.57	1.39E-05	212.81	2.88	36.14	35.39	2.11%
700.00	2.54E+07	156.98	7.01E-05	218.46	3.98	37.24	36.96	0.75%
800.00	3.54E+07	191.79	2.38E-04	223.51	5.11	38.36	38.32	0.11%
900.00	4.76E+07	228.85	6.20E-04	228.09	6.26	39.52	39.48	0.09%
1000.00	6.19E+07	268.04	1.35E-03	232.31	7.42	40.68	40.47	0.51%
1100.00	7.85E+07	309.23	2.56E-03	236.24	8.57	41.83	41.30	1.28%
1200.00	9.76E+07	352.34	4.41E-03	239.93	9.69	42.95	42.00	2.26%
1300.00	1.19E+08	397.29	7.03E-03	243.41	10.77	44.02	42.58	3.39%
1400.00	1.44E+08	444.00	1.06E-02	246.71	11.78	45.04	43.06	4.60%
1500.00	1.71E+08	492.41	1.51E-02	249.85	12.73	45.99	43.45	5.85%
1600.00	2.00E+08	542.47	2.09E-02	252.85	13.62	46.87	43.76	7.12%
1700.00	2.33E+08	594.11	2.79E-02	255.72	14.43	47.69	44.02	8.34%
1800.00	2.69E+08	647.30	3.63E-02	258.46	15.18	48.44	44.30	9.34%
1900.00	3.08E+08	701.98	4.61E-02	261.10	15.87	49.13	44.58	10.20%
2000.00	3.50E+08	758.12	5.76E-02	263.64	16.50	49.75	44.82	11.01%

Table 7: Thermodynamic Properties and Partition Functions for CO. Note: Constant values across all temperature vectors have been excluded from the core columns for clarity: $Q_{\text{el}} = 1$, $S_{\text{el}} = 0$, $C_{v,\text{trans}} = 12.47169$, $C_{v,\text{rot}} = 8.31446$.

T (K)	Q_{trans}	Q_{rot}	Q_{vib}	S_{tot}	$C_{v,\text{vib}}$	C_p	$C_{p,\text{NIST}}$	Err (%)
298.15	5.82E+06	109.01	0.0049	197.61	0.02	29.12	—	—
300.00	5.91E+06	109.68	0.0051	197.79	0.02	29.12	29.15	0.09%
400.00	1.21E+07	146.24	0.0190	206.19	0.19	29.29	29.15	0.48%
500.00	2.12E+07	182.80	0.0421	212.76	0.59	29.69	29.30	1.34%
600.00	3.34E+07	219.36	0.0717	218.23	1.19	30.29	29.82	1.58%
700.00	4.92E+07	255.92	0.1051	222.95	1.88	30.98	30.47	1.68%
800.00	6.87E+07	292.48	0.1406	227.13	2.58	31.68	31.17	1.64%
900.00	9.22E+07	329.04	0.1772	230.90	3.24	32.34	31.88	1.43%
1000.00	1.20E+08	365.61	0.2140	234.34	3.83	32.93	32.55	1.15%
1100.00	1.52E+08	402.17	0.2509	237.50	4.34	33.44	33.18	0.80%
1200.00	1.89E+08	438.73	0.2875	240.43	4.79	33.89	33.73	0.49%
1300.00	2.31E+08	475.29	0.3238	243.16	5.18	34.28	34.20	0.24%
1400.00	2.78E+08	511.85	0.3598	245.71	5.52	34.62	34.55	0.20%
1500.00	3.31E+08	548.41	0.3955	248.11	5.81	34.91	34.58	0.94%
1600.00	3.88E+08	584.97	0.4309	250.37	6.06	35.16	34.93	0.65%
1700.00	4.52E+08	621.53	0.4659	252.51	6.27	35.37	35.22	0.44%
1800.00	5.21E+08	658.09	0.5007	254.54	6.46	35.56	35.48	0.23%
1900.00	5.97E+08	694.65	0.5353	256.46	6.63	35.73	35.70	0.08%
2000.00	6.78E+08	731.21	0.5696	258.30	6.77	35.87	35.89	0.05%

Table 8: Thermodynamic Properties and Partition Functions for NO_2. Note: Constant values across all temperature vectors have been excluded from the core columns for clarity: $Q_{\text{el}} = 2$, $S_{\text{el}} = 5.762826$, $C_{v,\text{trans}} = C_{v,\text{rot}} = 12.47169$.

T (K)	Q_{trans}	Q_{rot}	Q_{vib}	S_{tot}	$C_{v,\text{vib}}$	C_p	$C_{p,\text{NIST}}$	Err (%)
298.15	1.23E+07	2249.82	9.41E-05	240.05	3.69	36.94	—	—
300.00	1.25E+07	2270.79	9.98E-05	240.27	3.74	37.00	33.57	10.21%
400.00	2.56E+07	3496.11	0.0011	251.33	6.77	40.03	38.26	4.62%
500.00	4.46E+07	4885.96	0.0045	260.58	9.72	42.98	41.74	2.97%
600.00	7.04E+07	6422.76	0.0124	268.65	12.32	45.57	44.09	3.36%
700.00	1.04E+08	8093.60	0.0265	275.85	14.46	47.72	45.60	4.65%
800.00	1.45E+08	9888.48	0.0482	282.34	16.20	49.46	46.51	6.34%
900.00	1.94E+08	11799.36	0.0788	288.24	17.59	50.85	47.02	8.14%
1000.00	2.53E+08	13819.58	0.1197	293.66	18.70	51.96	47.29	9.87%
1100.00	3.21E+08	15943.50	0.1718	298.66	19.60	52.85	47.42	11.46%
1200.00	3.98E+08	18166.30	0.2365	303.29	20.32	53.58	47.47	12.87%
1300.00	4.87E+08	20483.76	0.3148	307.60	20.92	54.18	47.46	14.15%
1400.00	5.86E+08	22892.16	0.4079	311.63	21.41	54.67	47.43	15.26%
1500.00	6.96E+08	25388.18	0.5168	315.42	21.82	55.08	47.38	16.25%
1600.00	8.18E+08	27968.85	0.6427	318.99	22.17	55.42	47.33	17.10%
1700.00	9.52E+08	30631.49	0.7866	322.35	22.46	55.72	47.29	17.82%
1800.00	1.10E+09	33373.63	0.9499	325.55	22.71	55.96	47.26	18.42%
1900.00	1.26E+09	36193.04	1.1334	328.58	22.92	56.18	47.24	18.92%
2000.00	1.43E+09	39087.66	1.3385	331.46	23.11	56.37	47.23	19.34%

4 Project 4: Quantum Chemical Energies, Basis Set Limits, and Partition Functions

4.1 Quantum Chemical based Calculation of energies and energy changes: effect of basis set

Table 9: Calculated Energies, Reaction Energies, and Relative Errors for the $H_2 \rightarrow 2H$ Dissociation Reaction across Various Levels of Theory and Basis Sets. The values for energies are in kcal/mol

Theory	**Basis Set**	$\mathbf{E_{H_2}}$	$\mathbf{E_H}$	**Energy**	**Expected Energy**	**Relative Error**
	Complete Basis Set	-737.66	-313.74	103.88	103.27	0.590%
CCSD(T)	aug-cc-pVTZ	-729.55	-313.64	102.26	103.27	0.97%
	cc-pVTZ	-729.35	-313.64	102.08	103.27	1.15%
	aug-cc-pVDZ	-724.78	-313.34	98.10	103.27	5.00%
	6-311+G(2d,p)	-726.82	-313.64	99.55	103.27	3.60%
	cc-pVDZ	-723.95	-313.30	97.35	103.27	5.73%
	6-31G(d,p)	-724.71	-312.65	99.42	103.27	3.73%
B2PLYP-D3	aug-cc-pVTZ	-727.89	-312.92	102.05	103.27	1.18%
	cc-pVTZ	-727.80	-312.89	102.03	103.27	1.20%
	aug-cc-pVDZ	-723.79	-312.59	98.61	103.27	4.51%
	6-311+G(2d,p)	-726.76	-312.88	100.99	103.27	2.21%
	cc-pVDZ	-723.41	-312.45	98.50	103.27	4.61%
	6-31G(d,p)	-725.75	-311.82	102.11	103.27	1.12%
MP2	aug-cc-pVTZ	-724.61	-313.64	97.32	103.27	5.76%
	cc-pVTZ	-724.36	-313.64	97.09	103.27	5.98%
	aug-cc-pVDZ	-719.16	-313.34	92.49	103.27	10.44%
	6-311+G(2d,p)	-721.60	-313.64	94.33	103.27	8.65%
	cc-pVDZ	-718.48	-313.30	91.87	103.27	11.04%
	6-31G(d,p)	-719.85	-312.65	94.56	103.27	8.43%
M06-2X	aug-cc-pVTZ	-727.10	-312.63	101.84	103.27	1.38%
	cc-pVTZ	-727.09	-312.58	101.92	103.27	1.30%
	aug-cc-pVDZ	-723.79	-312.44	98.91	103.27	4.21%
	6-311+G(2d,p)	-726.79	-312.58	101.62	103.27	1.59%
	cc-pVDZ	-723.60	-312.13	99.34	103.27	3.80%
	6-31G(d,p)	-726.02	-311.66	102.69	103.27	0.56%
B3LYP	aug-cc-pVTZ	-734.16	-315.17	103.81	103.27	0.53%
	cc-pVTZ	-734.14	-315.11	103.92	103.27	0.64%
	aug-cc-pVDZ	-730.48	-314.79	100.89	103.27	2.30%
	6-311+G(2d,p)	-733.87	-315.11	103.66	103.27	0.38%
	cc-pVDZ	-730.20	-314.54	101.11	103.27	2.09%
	6-31G(d,p)	-733.16	-313.93	105.31	103.27	1.98%
HF	aug-cc-pVTZ	-704.45	-313.64	77.16	103.27	25.28%
	cc-pVTZ	-704.40	-313.64	77.13	103.27	25.31%
	aug-cc-pVDZ	-701.84	-313.34	75.16	103.27	27.21%
	6-311+G(2d,p)	-704.08	-313.64	76.81	103.27	25.62%
	cc-pVDZ	-701.75	-313.30	75.15	103.27	27.23%
	6-31G(d,p)	-703.29	-312.65	78.00	103.27	24.47%

Table 10: Calculated Energies, Reaction Energies, and Relative Errors for the $CH_4 \rightarrow CH_3$+H Dissociation Reaction across Various Levels of Theory and Basis Sets. The values for energies are in kcal/mol

Theory	**Basis Set**	$\mathbf{E_{CH_4}}$	$\mathbf{E_{CH_3}}$	$\mathbf{E_H}$	**Energy**	**Expected Energy**	**Relative Error**
	Complete Basis Set	-25388.24	-24961.72	-313.75	103.31	103.36	0.046%
CCSD(T)	aug-cc-pVTZ	-25348.96	-24933.43	-313.64	101.88	103.36	1.43%
	cc-pVTZ	-25347.17	-24931.67	-313.64	101.87	103.36	1.45%
	aug-cc-pVDZ	-25320.92	-24909.06	-313.34	98.52	103.36	4.68%
	6-311+G(2d,p)	-25332.64	-24919.50	-313.64	99.50	103.36	3.73%
	cc-pVDZ	-25315.51	-24906.13	-313.30	96.07	103.36	7.05%
	6-31G(d,p)	-25315.97	-24903.29	-312.65	100.03	103.36	3.22%
B2PLYP-D3	aug-cc-pVTZ	-25378.48	-24963.96	-312.92	101.60	103.36	1.70%
	cc-pVTZ	-25377.84	-24963.14	-312.89	101.82	103.36	1.49%
	aug-cc-pVDZ	-25362.40	-24950.09	-312.59	99.72	103.36	3.52%
	6-311+G(2d,p)	-25372.05	-24958.38	-312.88	100.78	103.36	2.49%
	cc-pVDZ	-25359.15	-24946.61	-312.45	100.09	103.36	3.17%
	6-31G(d,p)	-25362.34	-24948.00	-311.82	102.52	103.36	0.81%
MP2	aug-cc-pVTZ	-25331.97	-24917.21	-313.64	101.12	103.36	2.17%
	cc-pVTZ	-25330.20	-24915.52	-313.64	101.04	103.36	2.24%
	aug-cc-pVDZ	-25302.90	-24882.11	-313.34	97.45	103.36	5.71%
	6-311+G(2d,p)	-25315.37	-24903.20	-313.64	98.53	103.36	4.67%
	cc-pVDZ	-25297.93	-24887.40	-313.30	97.22	103.36	5.94%
	6-31G(d,p)	-25299.97	-24888.28	-312.65	99.04	103.36	4.18%
M06-2X	aug-cc-pVTZ	-25387.01	-24972.14	-312.63	102.24	103.36	1.08%
	cc-pVTZ	-25386.79	-24971.74	-312.58	102.47	103.36	0.86%
	aug-cc-pVDZ	-25376.29	-24963.02	-312.44	100.83	103.36	2.45%
	6-311+G(2d,p)	-25384.52	-24969.86	-312.58	102.08	103.36	1.24%
	cc-pVDZ	-25374.25	-24960.33	-312.13	101.79	103.36	1.52%
	6-31G(d,p)	-25377.19	-24961.29	-311.66	104.24	103.36	0.85%
B3LYP	aug-cc-pVTZ	-25410.30	-24993.53	-315.17	101.60	103.36	1.71%
	cc-pVTZ	-25410.16	-24993.08	-315.11	101.96	103.36	1.35%
	aug-cc-pVDZ	-25399.33	-24984.12	-314.79	100.41	103.36	2.85%
	6-311+G(2d,p)	-25408.09	-24991.48	-315.11	101.51	103.36	1.79%
	cc-pVDZ	-25396.58	-24980.96	-314.54	101.07	103.36	2.21%
	6-31G(d,p)	-25400.95	-24983.12	-313.93	103.91	103.36	0.53%
HF	aug-cc-pVTZ	-25205.04	-24816.34	-313.64	75.06	103.36	27.38%
	cc-pVTZ	-25204.89	-24816.07	-313.64	75.19	103.36	27.26%
	aug-cc-pVDZ	-25196.37	-24808.64	-313.34	74.40	103.36	28.02%
	6-311+G(2d,p)	-25202.93	-24814.35	-313.64	74.95	103.36	27.49%
	cc-pVDZ	-25195.60	-24807.48	-313.30	74.83	103.36	27.60%
	6-31G(d,p)	-25197.18	-24807.72	-312.65	76.81	103.36	25.68%

Theory	Basis Set	$E_{CH_3NH_2}$	E_{CH_3}	E_{NH_2}	Energy	Expected Energy	Relative Error
	Complete Basis Set	-60079.54	-24961.71704	-35026.88747	81.32	82.51649	1.455%
CCSD(T)	aug-cc-pVTZ	-60031.60	-24933.43	-35002.99	95.18	82.52	15.34%
	cc-pVTZ	-60025.57	-24931.67	-34999.05	94.85	82.52	14.94%
	aug-cc-pVDZ	-59972.76	-24909.06	-34973.07	90.63	82.52	9.83%
	6-311+G(2d,p)	-59981.15	-24919.50	-34984.98	76.66	82.52	7.09%
	cc-pVDZ	-59940.94	-24906.13	-34961.16	73.65	82.52	10.74%
	6-31G(d,p)	-59940.07	-24903.29	-34959.78	77.01	82.52	6.68%
B2PLYP-D3	aug-cc-pVTZ	-60083.63	-24963.96	-35039.96	79.71	82.52	3.40%
	cc-pVTZ	-60080.23	-24963.14	-35037.25	79.85	82.52	3.24%
	aug-cc-pVDZ	-60053.70	-24950.09	-35024.45	79.16	82.52	4.07%
	6-311+G(2d,p)	-60070.07	-24958.38	-35032.73	78.95	82.52	4.32%
	cc-pVDZ	-60041.83	-24946.61	-35015.29	79.93	82.52	3.14%
	6-31G(d,p)	-60043.67	-24948.00	-35015.21	80.45	82.52	2.50%
MP2	aug-cc-pVTZ	-59991.45	-24917.21	-34988.17	86.07	82.52	4.31%
	cc-pVTZ	-59985.37	-24915.52	-34984.34	85.50	82.52	3.62%
	aug-cc-pVDZ	-59932.94	-24882.11	-34958.04	82.78	82.52	0.32%
	6-311+G(2d,p)	-59956.84	-24903.20	-34970.62	83.02	82.52	0.61%
	cc-pVDZ	-59916.22	-24887.40	-34946.89	81.93	82.52	0.71%
	6-31G(d,p)	-59917.13	-24888.28	-34946.15	82.70	82.52	0.23%
M06-2X	aug-cc-pVTZ	-60103.54	-24972.14	-35048.12	83.28	82.52	0.93%
	cc-pVTZ	-60101.68	-24971.74	-35046.38	83.56	82.52	1.27%
	aug-cc-pVDZ	-60085.80	-24963.02	-35038.64	84.14	82.52	1.97%
	6-311+G(2d,p)	-60098.17	-24969.86	-35044.95	83.36	82.52	1.02%
	cc-pVDZ	-60077.44	-24960.33	-35031.31	85.79	82.52	3.97%
	6-31G(d,p)	-60077.39	-24961.29	-35030.17	85.92	82.52	4.13%
B3LYP	aug-cc-pVTZ	-60140.31	-24993.53	-35070.01	76.77	82.52	6.97%
	cc-pVTZ	-60138.13	-24993.08	-35067.84	77.21	82.52	6.44%
	aug-cc-pVDZ	-60121.91	-24984.12	-35060.34	77.45	82.52	6.14%
	6-311+G(2d,p)	-60135.51	-24991.48	-35067.16	76.88	82.52	6.83%
	cc-pVDZ	-60111.96	-24980.96	-35051.99	79.02	82.52	4.24%
	6-31G(d,p)	-60115.09	-24983.12	-35052.69	79.28	82.52	3.93%
HF	aug-cc-pVTZ	-59731.51	-24816.34	-34869.50	45.67	82.52	44.66%
	cc-pVTZ	-59730.02	-24816.07	-34868.20	45.75	82.52	44.56%
	aug-cc-pVDZ	-59715.77	-24808.64	-34861.24	45.89	82.52	44.38%
	6-311+G(2d,p)	-59726.47	-24814.35	-34866.59	45.53	82.52	44.83%
	cc-pVDZ	-59709.83	-24807.48	-34856.16	46.20	82.52	44.02%
	6-31G(d,p)	-59709.64	-24807.72	-34854.61	47.32	82.52	42.66%

4.2 Calculation of Basis and Primitive Gaussian Functions

To determine the number of contracted basis functions (BF) and primitive Gaussian functions (PG) for H_2, CH_4, and CH_3NH_2, we evaluate the atomic contributions for Hydrogen (H), Carbon (C), and Nitrogen (N) under three different basis sets: STO-3G, 6-31G(d), and cc-pVDZ.

Atomic Contributions

- **STO-3G:** 3 GTOs to simulate a single STO
  - H ($1s$): 1 BF, 3 PGs.
  - C / N ($1s$, $2s$, $2p$): 5 BFs, 15 PGs.
- **6-31G(d):** Each basis function is made of 6 primitve GTOs in core orbitals, valence orbitals are "split" into two basis functions. The inner valence part is a contraction of 3 primitives, and the outer valence part is described by 1 primitive. Further more, a single set of d-type polarization functions is added to heavy atoms
  - H ($1s$-valence): No core, so 2 BFs, 4 PGs.
  - C / N ($1s$-core, $2s$-valence, $2p$-valence, $3d$-polarization):

$$1 + 4 + 4 + 6 = 15 \text{ BFs}; \; 6 + 3 \cdot 4 + 4 + 6 = 28 \text{ PGs}.$$

- **cc-pVDZ:** Doubles the number of primitive functions used to describe that valence orbital and polarization functions are included. This basis-set use 5 spherical d-functions. The basis set is generated by contracting primitive Gaussian functions: (PG $\rightarrow$ [BF]) (i.e$(4s, 1p) \rightarrow [2s, 1p]$ for H, and $(9s, 4p, 1d) \rightarrow [3s, 2p, 1d]$ for heavy atoms).

– H ($2s$-valence, $1p$-polarization):

$$2 + 3 = 5 \text{ BFs}; \; 4 + 3 = 7 \text{ PGs}.$$

– C / N ($1s$-core, $2s$-valence, $2p$-valence, $1d$-polarization):

$$1 + 2 + 3 + 3 + 5 = 14 \text{ BFs}; \; 9 + 4 \cdot 3 + 5 = 26 \text{ PGs}.$$

System Calculations

- **System 1:** (H_2)
  - **STO-3G:** $2 \times 1 = 2$ BFs; $2 \times 3 = 6$ PGs
  - **6-31G(d):** $2 \times 2 = 4$ BFs; $2 \times 4 = 8$ PGs
  - **cc-pVDZ:** $2 \times 5 = 10$ BFs; $2 \times 7 = 14$ PGs
- **System 2: Methane (CH_4)**
  - **STO-3G:** 5 (C) + 4×1 (H) = 9 BFs; 15 (C) + 4×3 (H) = 27 PGs
  - **6-31G(d):** 15 (C) + 4×2 (H) = 23 BFs; 28 (C) + 4×4 (H) = 44 PGs
  - **cc-pVDZ:** 14 (C) + 4×5 (H) = 34 BFs; 26 (C) + 4×7 (H) = 54 PGs
- **System 3: Methylamine (CH_3NH_2)**
  - **STO-3G:** 5 (C) + 5 (N) + 5×1 (H) = 15 BFs; 15 (C) + 15 (N) + 5×3 (H) = 45 PGs
  - **6-31G(d):** 15 (C) + 15 (N) + 5×2 (H) = 40 BFs; 28 (C) + 28 (N) + 5×4 (H) = 76 PGs
  - **cc-pVDZ:** 14 (C) + 14 (N) + 5×5 (H) = 53 BFs; 26 (C) + 26 (N) + 5×7 (H) = 87 PGs

4.3 Estimation of the equilibrium constant from the molecular partition function

The equilibrium constant of a chemical reaction can be calculated directly from the molecular partition functions of the reactants and products using statistical mechanics. For a general reaction, the equilibrium constant K is given by the ratio of the product partition functions to the reactant partition functions, multiplied by an exponential factor accounting for the zero-point energy difference:

$$K = \frac{Q_{tot}^{PROD}}{Q_{tot}^{REACT}} \exp\left(-\frac{\Delta\varepsilon_0}{k_B T}\right)$$

Table 11: Transposed Partition Functions and Equilibrium Constants for $CH_4 \leftrightarrow CH_3 + H$

	Methane (CH_4)				**Methyl Radical (CH_3)**				**Hydrogen Atom (H)**					
T (K)	$\mathbf{Q}_{\text{trans}}$	$\mathbf{Q}_{\text{rot}}$	$\mathbf{Q}_{\text{vib}}$	$\mathbf{Q}_{\text{elec}}$	$\mathbf{Q}_{\text{trans}}$	$\mathbf{Q}_{\text{rot}}$	$\mathbf{Q}_{\text{vib}}$	$\mathbf{Q}_{\text{elec}}$	$\mathbf{Q}_{\text{trans}}$	$\mathbf{Q}_{\text{rot}}$	$\mathbf{Q}_{\text{vib}}$	$\mathbf{Q}_{\text{elec}}$	$\mathbf{K}_{\text{fwd}}$	$\mathbf{K}_{\text{rev}}$
298.15	2.52E+06	36.51	1.01	1	2.29E+06	127.32	1.12	2	3.98E+04	1	1	2	2.85E-71	3.51E+70
300	2.56E+06	36.85	1.01	1	2.32E+06	128.50	1.12	2	4.04E+04	1	1	2	8.57E-71	1.17E+70
400	5.26E+06	56.74	1.03	1	4.77E+06	197.84	1.24	2	8.29E+04	1	1	2	1.72E-51	5.81E+50
500	9.19E+06	79.30	1.09	1	8.34E+06	276.49	1.40	2	1.45E+05	1	1	2	7.58E-40	1.32E+39
600	1.45E+07	104.24	1.18	1	1.31E+07	363.46	1.59	2	2.28E+05	1	1	2	4.77E-32	2.10E+31
700	2.13E+07	131.36	1.32	1	1.93E+07	458.01	1.82	2	3.36E+05	1	1	2	1.87E-26	5.34E+25
800	2.98E+07	160.49	1.50	1	2.70E+07	559.58	2.09	2	4.69E+05	1	1	2	3.03E-22	3.30E+21
900	3.99E+07	191.50	1.75	1	3.62E+07	667.72	2.40	2	6.30E+05	1	1	2	5.83E-19	1.72E+18
1000	5.20E+07	224.29	2.06	1	4.72E+07	782.04	2.78	2	8.19E+05	1	1	2	2.50E-16	3.99E+15
1100	6.60E+07	258.76	2.45	1	5.98E+07	902.24	3.21	2	1.04E+06	1	1	2	3.61E-14	2.77E+13
1200	8.20E+07	294.84	2.94	1	7.44E+07	1028.02	3.71	2	1.29E+06	1	1	2	2.28E-12	4.38E+11
1300	1.00E+08	332.45	3.55	1	9.09E+07	1159.17	4.29	2	1.58E+06	1	1	2	7.68E-11	1.30E+10
1400	1.21E+08	371.54	4.30	1	1.09E+08	1295.46	4.96	2	1.90E+06	1	1	2	1.57E-09	6.38E+08
1500	1.43E+08	412.05	5.23	1	1.30E+08	1436.71	5.74	2	2.26E+06	1	1	2	2.15E-08	4.66E+07
1600	1.68E+08	453.93	6.36	1	1.53E+08	1582.75	6.62	2	2.65E+06	1	1	2	2.12E-07	4.72E+06
1700	1.96E+08	497.15	7.75	1	1.78E+08	1733.42	7.64	2	3.09E+06	1	1	2	1.60E-06	6.24E+05
1800	2.26E+08	541.65	9.45	1	2.05E+08	1888.60	8.80	2	3.56E+06	1	1	2	9.68E-06	1.03E+05
1900	2.59E+08	587.41	11.50	1	2.35E+08	2048.15	10.11	2	4.08E+06	1	1	2	4.84E-05	2.07E+04
2000	2.94E+08	634.39	13.99	1	2.67E+08	2211.95	11.61	2	4.63E+06	1	1	2	2.06E-04	4.85E+03

5 Project 5: Determination of rate constants of abstraction reactions using transition state theory

To compute the standard Transition State Theory (TST) rate constant (k_{TST}) using calculating partition functions for a general bimolecular abstraction reaction (e.g., $A + B \rightarrow \mathrm{TS}^{\ddagger} \rightarrow$ Products), the following equation is required:

$$k_{TST} = \frac{k_B T}{h} \frac{q^{\ddagger}}{q_A q_B} \exp\left(-\frac{\Delta E_0^{\ddagger}}{k_B T}\right)$$

. Here $q^{\ddagger}$ is the total molecular partition function of the transition state

To account for light atoms that tunnel through the barrier due to their quantum mechanical wave nature, a multiplicative transmission coefficient, $\kappa(T)$, is introduced to the standard TST rate constant equation:

$$k_{TST/tunneling} = \kappa(T) \cdot k_{TST}$$

There are a lot of way to compute $\kappa(T)$, here I use Wigner Correction. The Wigner correction assumes a parabolic barrier top and only requires the imaginary frequency ($\nu^{\ddagger}$) of the transition state. It is relatively accurate at high temperatures but underestimates tunnelling at low temperatures.

$$\kappa_{Wigner}(T) = 1 + \frac{1}{24}\left(\frac{h|\nu^{\ddagger}|}{k_B T}\right)^2$$

Table 12: Partition Functions and Rate Constants for $CH_4 + H \leftrightarrow TS \rightarrow$ Products

	Methane (CH_4)				**Hydrogen Atom (H)**				**Transition State (TS)**				**Rate Constants**		
T (K)	$\mathbf{Q}_{\mathrm{trans}}$ [-]	$\mathbf{Q}_{\mathrm{rot}}$ [-]	$\mathbf{Q}_{\mathrm{vib}}$ [-]	$\mathbf{g}_{\mathrm{el}}$ [-]	$\mathbf{Q}_{\mathrm{trans}}$ [-]	$\mathbf{Q}_{\mathrm{rot}}$ [-]	$\mathbf{Q}_{\mathrm{vib}}$ [-]	$\mathbf{g}_{\mathrm{el}}$ [-]	$\mathbf{Q}_{\mathrm{trans}}$ [-]	$\mathbf{Q}_{\mathrm{rot}}$ [-]	$\mathbf{Q}_{\mathrm{vib}}$ [-]	$\mathbf{g}_{\mathrm{el}}$ [-]	$\mathbf{k_b}$ ($\mathbf{m^3\,s^{-1}}$)	$\mathbf{k_{b,corr}}$ ($\mathbf{m^3\,s^{-1}}$)	$\mathbf{k_{NIST}}$ ($\mathbf{m^3\,s^{-1}}$)
298.15	2.52E+06	36.02	1.01	1.00	3.98E+04	1.00	1.00	2.00	2.76E+06	1202.20	1.19	2.00	3.93E-26	1.38E-25	8.14E-26
300.00	2.56E+06	36.36	1.01	1.00	4.04E+04	1.00	1.00	2.00	2.81E+06	1213.41	1.20	2.00	4.52E-26	1.59E-25	9.27E-26
400.00	5.26E+06	55.98	1.03	1.00	8.29E+04	1.00	1.00	2.00	5.76E+06	1868.16	1.48	2.00	1.32E-23	3.20E-23	1.32E-23
500.00	9.19E+06	78.23	1.09	1.00	1.45E+05	1.00	1.00	2.00	1.01E+07	2610.84	1.92	2.00	4.26E-22	8.22E-22	2.95E-22
600.00	1.45E+07	102.84	1.19	1.00	2.29E+05	1.00	1.00	2.00	1.59E+07	3432.03	2.60	2.00	4.59E-21	7.60E-21	2.55E-21
700.00	2.13E+07	129.59	1.33	1.00	3.36E+05	1.00	1.00	2.00	2.33E+07	4324.86	3.60	2.00	2.63E-20	3.91E-20	1.27E-20
800.00	2.98E+07	158.33	1.52	1.00	4.69E+05	1.00	1.00	2.00	3.26E+07	5283.96	5.07	2.00	1.01E-19	1.39E-19	4.42E-20
900.00	3.99E+07	188.92	1.77	1.00	6.30E+05	1.00	1.00	2.00	4.38E+07	6305.05	7.18	2.00	2.97E-19	3.86E-19	1.21E-19
1000.00	5.20E+07	221.27	2.09	1.00	8.19E+05	1.00	1.00	2.00	5.70E+07	7384.56	10.20	2.00	7.18E-19	8.96E-19	2.79E-19
1100.00	6.60E+07	255.27	2.49	1.00	1.04E+06	1.00	1.00	2.00	7.23E+07	8519.49	14.49	2.00	1.51E-18	1.82E-18	5.66E-19
1200.00	8.20E+07	290.86	2.99	1.00	1.29E+06	1.00	1.00	2.00	8.98E+07	9707.25	20.54	2.00	2.84E-18	3.35E-18	1.04E-18
1300.00	1.00E+08	327.97	3.61	1.00	1.58E+06	1.00	1.00	2.00	1.10E+08	10945.60	29.01	2.00	4.93E-18	5.68E-18	1.77E-18
1400.00	1.21E+08	366.53	4.38	1.00	1.90E+06	1.00	1.00	2.00	1.32E+08	12232.54	40.79	2.00	7.98E-18	9.03E-18	2.83E-18
1500.00	1.43E+08	406.49	5.33	1.00	2.26E+06	1.00	1.00	2.00	1.57E+08	13566.30	57.03	2.00	1.22E-17	1.36E-17	4.31E-18
1600.00	1.68E+08	447.81	6.50	1.00	2.65E+06	1.00	1.00	2.00	1.84E+08	14945.30	79.28	2.00	1.79E-17	1.97E-17	6.28E-18
1700.00	1.96E+08	490.45	7.92	1.00	3.09E+06	1.00	1.00	2.00	2.15E+08	16368.09	109.54	2.00	2.52E-17	2.70E-17	8.85E-18
1800.00	2.26E+08	534.35	9.66	1.00	3.56E+06	1.00	1.00	2.00	2.48E+08	17833.36	150.38	2.00	3.44E-17	3.66E-17	1.21E-17
1900.00	2.59E+08	579.49	11.77	1.00	4.08E+06	1.00	1.00	2.00	2.83E+08	19339.93	205.14	2.00	4.57E-17	4.83E-17	1.62E-17
2000.00	2.94E+08	625.84	14.32	1.00	4.64E+06	1.00	1.00	2.00	3.22E+08	20886.68	278.05	2.00	5.93E-17	6.23E-17	2.11E-17

Table 13: Partition Functions and Rate Constants for $CH_3 + H_2 \leftrightarrow TS \rightarrow$ Products

	Methyl Radical (CH_3)				Hydrogen Molecule (H_2)				Transition State (TS)				Rate Constants		
T (K)	Q_{trans} [-]	Q_{rot} [-]	Q_{vib} [-]	g_{el} [-]	Q_{trans} [-]	Q_{rot} [-]	Q_{vib} [-]	g_{el} [-]	Q_{trans} [-]	Q_{rot} [-]	Q_{vib} [-]	g_{el} [-]	k_b ($m^3 s^{-1}$)	$k_{b,corr}$ ($m^3 s^{-1}$)	k_{NIST} ($m^3 s^{-1}$)
298.15	2.29E+06	125.33	1.15	2.00	1.12E+05	1.69	1.00	1.00	2.76E+06	1202.20	1.19	2.00	1.34E-27	4.71E-27	1.66E-26
300.00	2.32E+06	126.50	1.15	2.00	1.14E+05	1.70	1.00	1.00	2.81E+06	1213.41	1.20	2.00	1.53E-27	5.39E-27	1.86E-26
400.00	4.77E+06	194.76	1.30	2.00	2.34E+05	2.27	1.00	1.00	5.76E+06	1868.16	1.48	2.00	3.49E-25	8.48E-25	1.65E-24
500.00	8.34E+06	272.19	1.47	2.00	4.10E+05	2.84	1.00	1.00	1.01E+07	2610.84	1.92	2.00	9.09E-24	1.75E-23	2.80E-23
600.00	1.31E+07	357.80	1.68	2.00	6.46E+05	3.40	1.00	1.00	1.59E+07	3432.03	2.60	2.00	8.20E-23	1.36E-22	2.02E-22
700.00	1.93E+07	450.88	1.94	2.00	9.50E+05	3.97	1.00	1.00	2.33E+07	4324.86	3.60	2.00	4.07E-22	6.05E-22	8.82E-22
800.00	2.70E+07	550.86	2.23	2.00	1.33E+06	4.54	1.00	1.00	3.26E+07	5283.96	5.07	2.00	1.40E-21	1.92E-21	2.80E-21
900.00	3.62E+07	657.31	2.58	2.00	1.78E+06	5.11	1.00	1.00	4.38E+07	6305.05	7.18	2.00	3.74E-21	4.87E-21	7.14E-21
1000.00	4.72E+07	769.86	2.99	2.00	2.32E+06	5.67	1.00	1.00	5.70E+07	7384.56	10.20	2.00	8.45E-21	1.05E-20	1.56E-20
1100.00	5.98E+07	888.17	3.46	2.00	2.94E+06	6.24	1.00	1.00	7.23E+07	8519.49	14.49	2.00	1.68E-20	2.03E-20	3.02E-20
1200.00	7.44E+07	1012.00	4.02	2.00	3.66E+06	6.81	1.00	1.00	8.98E+07	9707.25	20.54	2.00	3.04E-20	3.58E-20	5.35E-20
1300.00	9.09E+07	1141.10	4.65	2.00	4.47E+06	7.38	1.01	1.00	1.10E+08	10945.60	29.01	2.00	5.11E-20	5.88E-20	8.83E-20
1400.00	1.09E+08	1275.27	5.39	2.00	5.37E+06	7.94	1.01	1.00	1.32E+08	12232.54	40.79	2.00	8.09E-20	9.16E-20	1.38E-19
1500.00	1.30E+08	1414.32	6.24	2.00	6.39E+06	8.51	1.01	1.00	1.57E+08	13566.30	57.03	2.00	1.22E-19	1.36E-19	2.05E-19
1600.00	1.53E+08	1558.08	7.21	2.00	7.50E+06	9.08	1.02	1.00	1.84E+08	14945.30	79.28	2.00	1.77E-19	1.95E-19	2.94E-19
1700.00	1.78E+08	1706.41	8.33	2.00	8.73E+06	9.65	1.02	1.00	2.15E+08	16368.09	109.54	2.00	2.48E-19	2.66E-19	4.08E-19
1800.00	2.05E+08	1859.17	9.60	2.00	1.01E+07	10.21	1.03	1.00	2.48E+08	17833.36	150.38	2.00	3.38E-19	3.60E-19	5.50E-19
1900.00	2.35E+08	2016.23	11.04	2.00	1.15E+07	10.78	1.04	1.00	2.83E+08	19339.93	205.14	2.00	4.50E-19	4.76E-19	7.25E-19
2000.00	2.67E+08	2177.48	12.69	2.00	1.31E+07	11.35	1.04	1.00	3.22E+08	20886.68	278.05	2.00	5.86E-19	6.16E-19	9.37E-19

6 Project 6: First-Principles NMR Chemical Shift Estimation and Thermochemical Analysis

This report outlines the first-principles calculation of the isotropic shielding (σ_{iso}) and J-couplings for ethanol (CH_3CH_2OH).

All computations were carried out utilizing `Gaussian16`. The molecular geometry of ethanol was optimized using

```
# b3lyp/6-311+g(d,p) opt freq
```

Subsequently, a single-point NMR calculation using the GIAO method was performed on the optimized structure. The tetramethylsilane (TMS) standard was calculated utilizing the exact same theoretical method and basis set.

```
# b3lyp/6-311+g(d,p) Freq nmr=giao int=ultrafine
```

Quantum chemistry initially calculates the absolute magnetic shielding, σ_{iso}. The chemical shift (δ), expressed in parts per million (ppm), is then defined relative to the standard TMS reference:

$$\delta = \sigma_{iso}^{TMS} - \sigma_{iso}^{sample} \tag{51}$$

The computed isotropic shielding for the 12 equivalent protons of TMS was determined to be $\sigma_{iso}^{TMS} = 32.7014$ ppm. Table 14 compares the calculated data for the distinct proton environments in ethanol to experimental values reported in ChemicalBook.

Table 14: Computed isotropic shieldings, calculated chemical shifts, and experimental chemical shifts for Ethanol.

Functional Group	Computed σ_{iso} (ppm)	Computed δ (ppm)	ChemicalBook δ (ppm)
-CH_2- (Group A)	28.2107	4.4907	3.6870
-CH_3 (Group B)	30.7845	1.9169	1.2260
-OH (Group C)	32.2440	0.4574	2.6100

A major deviation occurs, beacuse in a gas-phase calculation at 0 K, the molecule is isolated; whereas, in experimental liquid NMR, rapid intermolecular hydrogen exchange heavily deshields the proton.

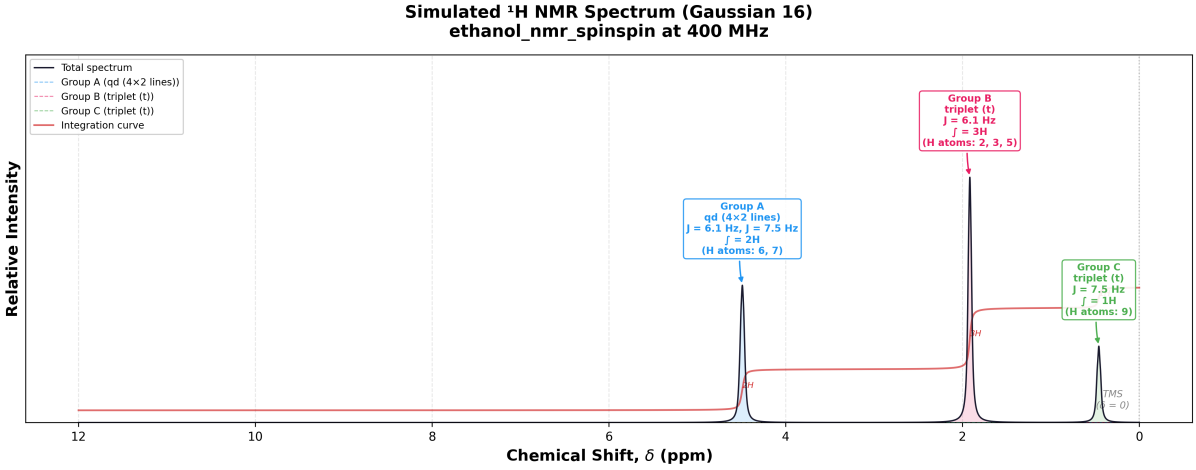
The spin-spin Splitting and J-coupling were calculated using the following command on `Gaussian16`

```
# b3lyp/6-311+g(d,p) Freq nmr=(giao, spinspin) int=ultrafine
```

The underlying theoretical vicinal coupling constant was extracted at approximately $^3J_{HH} = 6.09$ Hz, closely mirroring the typical experimental range (≈ 7.0 Hz) for ethanol.

Figure 3 shows the obtained ^{1}H NMR spectrum from the simulations. The spin-spin splitting patterns are as follows:

- **Group B (-CH3):** A triplet (coupled to the 2 protons on the -CH2- group).
- **Group C (-OH):** A triplet (coupled to the 2 protons on the -CH2- group). While experimental hydroxyl protons are typically highly deshielded due to intermolecular hydrogen bonding, the exceptionally low chemical shift here reflects an isolated gas-phase simulation where the proton remains highly shielded.
- **Group A (-CH2-):** A complex multiplet, identified as a "qd" (quartet of doublets), because it is split by both the 3 -CH_3 protons and the 1 -OH proton. The chemical shift is high because this carbon is directly attached to the highly electronegative oxygen atom, which deshields the protons by withdrawing electron density.



Group	δ (ppm)	Pattern	J (Hz)	# H
Group A	4.49	qd	6.1, 7.5	2
Group B	1.92	triplet	6.1	3
Group C	0.46	triplet	7.5	1

Figure 3: Simulated ^{1}H NMR spectrum of ethanol with spin-spin coupling computed at the B3LYP/6-311+G(d,p) level.

7 Project 7: Computational Chemical Kinetics and Transition State Theory

This section details the kinetic analysis of three distinct reactions, evaluating classical Transition State Theory (TST) alongside quantum tunneling corrections (Wigner and Eckart models). I used the M4 configuration: so $\omega B97X - D$ functional and `def2TZVP` as basis set and the `eyringpy` framework.

7.1 Reaction 1: $CH_4 + OH \rightarrow CH_3 + OH$

The Arrhenius plot for $CH_4 + OH \rightarrow CH_3 + OH$ in Figure 4 displays a noticeable upward curvature at lower temperatures, which is indicative of quantum mechanical tunneling.

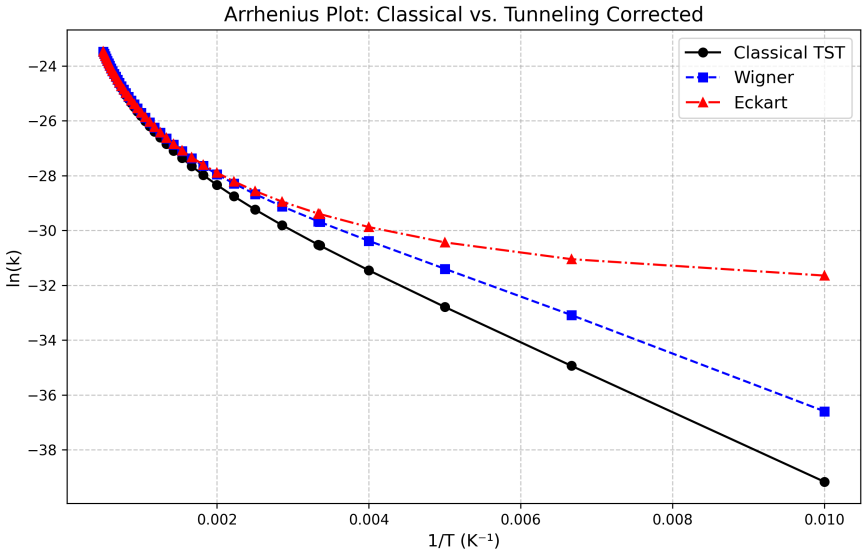


Figure 4: Arrhenius plot for Reaction 1 comparing classical TST, Wigner, and Eckart tunneling corrections.

The rate constants and tunneling enhancement factors evaluated at 300 K demonstrate the impact of the corrections:

- The classical rate is calculated as $5.579 \cdot 10^{-14}$.
- The Wigner rate is $1.307 \cdot 10^{-13}$, providing an enhancement factor of 2.34.
- The Eckart rate is $1.736 \cdot 10^{-13}$, representing an enhancement factor of 3.11.

For temperatures where tunneling is negligible (1000 K to 2000 K), the apparent Arrhenius parameters were extracted with an excellent linear fit ($R^2 = 0.9931$). The fitted activation energy ($E_a = -R \cdot \text{slope}$) is 38.95 kJ mol^{-1}, and the pre-exponential factor ($A = e^{\text{intercept}}$) is 5.830e-10.

A thermodynamic bottleneck analysis at 300 K reveals the following properties:

- Enthalpy of Activation ($\Delta H^{\ddagger}$, extracted from `eyringpy`): 7.11 kJ/mol
- Entropic Term ($-T\Delta S^{\ddagger}$, extracted from `eyringpy`): 24.98 kJ/mol
- Gibbs Free Energy ($\Delta G^{\ddagger} = \Delta H^{\ddagger} - T\Delta S^{\ddagger}$): 32.09 kJ/mol

The conclusion is that the bottleneck is entropic as $|-T\Delta S^{\ddagger}| \geq |\Delta H^{\ddagger}|$.

7.2 Reaction 2: $HNC \rightarrow HCN$

The Arrhenius plot for Reaction 2: $HNC \rightarrow HCN$ also illustrates the deviation introduced by the tunneling corrections at the lower end of the temperature scale.

At the target temperature of 300 K, the kinetics are defined as follows:

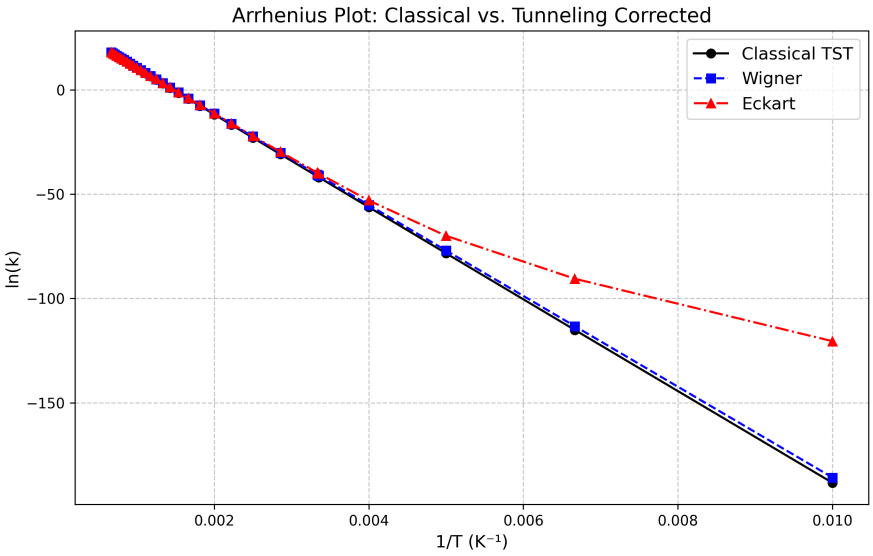


Figure 5: Arrhenius plot for Reaction 2 comparing classical TST, Wigner, and Eckart tunneling corrections.

- The classical rate is $1.001 \cdot 10^{-18}$.
- The Wigner rate is $2.273 \cdot 10^{-18}$, resulting in an enhancement factor of 2.27.
- The Eckart rate is $5.757 \cdot 10^{-18}$, showing a stronger enhancement factor of 5.75.

The apparent Arrhenius parameters, fitted between 1000 K and 1500 K ($R^2 = 1.0000$), show a much higher activation energy (E_a) of 184.96 kJ/mol and a pre-exponential factor (A) of $1.658 \cdot 10^{14}$.

A thermodynamic bottleneck analysis at 300 K reveals the following properties:

- Enthalpy of Activation ($\Delta H^\ddagger$, extracted from `eyringpy`): 182.00 kJ/mol
- Entropic Term ($-T\Delta S^\ddagger$, extracted from `eyringpy`): -5.15 kJ/mol
- Gibbs Free Energy ($\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger$): 176.86 kJ/mol

The conclusion is that the bottleneck is enthalpic as $|\Delta H^\ddagger| > |-T\Delta S^\ddagger|$

7.3 Reaction 3: $cis - HCOOH \rightarrow trans - HCOOH$

The final evaluation focuses on Reaction 3: $cis - HCOOH \rightarrow trans - HCOOH$, where the generated Arrhenius plot visualizes the tunneling corrections against the classical TST baseline.

The specific rates and enhancements at 300 K are computed as:

- The classical rate is $1.253 \cdot 10^{-4}$.
- The Wigner rate is $1.733 \cdot 10^{-4}$, which gives a modest enhancement factor of 1.38.
- The Eckart rate is $1.906 \cdot 10^{-4}$, yielding an enhancement factor of 1.52.

The Arrhenius fit for the 1000 K to 1500 K range is perfectly linear ($R^2 = 1.0000$), establishing an activation energy (E_a) of 54.35 kJ/mol and a pre-exponential factor (A) of $2.446 \cdot 10^{13}$.

A thermodynamic bottleneck analysis at 300 K reveals the following properties:

- Enthalpy of Activation ($\Delta H^\ddagger$, extracted from `eyringpy`): 49.79 kJ/mol
- Entropic Term ($-T\Delta S^\ddagger$, extracted from `eyringpy`): 0.38 kJ/mol
- Gibbs Free Energy ($\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger$): 50.17 kJ/mol

The conclusion is that the bottleneck is enthalpic as $|\Delta H^\ddagger| > |-T\Delta S^\ddagger|$

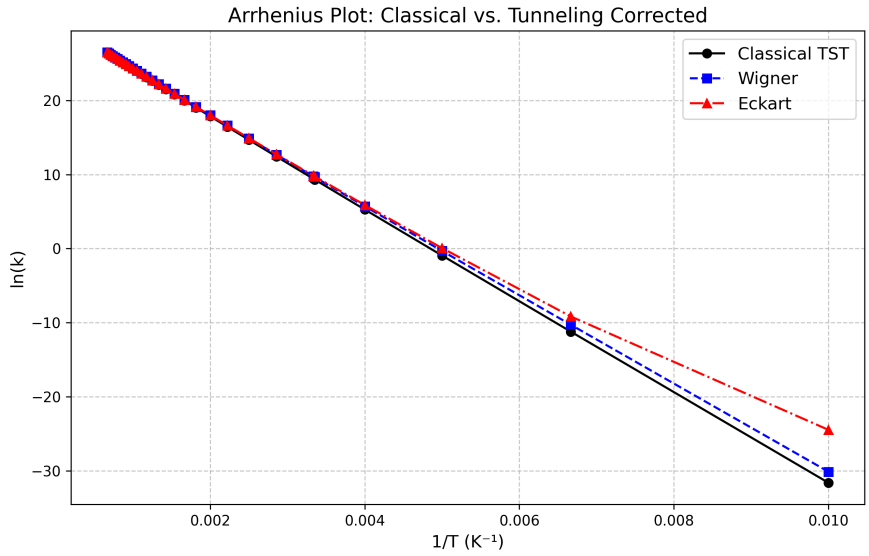


Figure 6: Arrhenius plot for Reaction 3 comparing classical TST, Wigner, and Eckart tunneling corrections.

7.4 Answers to Discussion Questions

1. **Which reaction has the larger classical activation free energy?**
   The $HNC \rightarrow HCN$ isomerization has a significantly larger classical activation free energy of value 184.96 kJ/mol.

2. **Which reaction has the larger imaginary transition-state frequency?**
   The $HNC \rightarrow HCN$ reaction has the larger imaginary transition-state frequency of value $-1152.1481\,\mathrm{cm}^{-1}$.

3. **Which reaction shows the larger Wigner correction at 300 K?**
   The $HNC \rightarrow HCN$ reaction shows the larger Wigner correction. The Wigner equation is directly proportional to the square of the imaginary frequency ($\kappa_{Wigner} \propto |\nu^{\ddagger}|^2$). Therefore, the larger imaginary frequency of the HNC $\rightarrow$ HCN transition state mathematically dictates a higher Wigner transmission coefficient.

4. **Which reaction shows the larger Eckart correction at 300 K?**
   The $HNC \rightarrow HCN$ reaction shows the substantially larger Eckart correction of value 5.75. Because the Eckart model maps the full one-dimensional barrier profile, it explicitly captures the massive quantum mechanical probability of a light hydrogen atom tunneling through a tall, narrow barrier at low temperatures.

5. **Does the Eckart correction differ significantly from the Wigner correction?**
   Yes, it differs significantly, particularly for the $HNC \rightarrow HCN$ reaction. That's beacuse Wigner correction is a simple first-order parabolic approximation, while Eckart correction incorporates the full asymmetric shape of the barrier (using forward and reverse reaction energies), yielding a much larger and more physically realistic tunneling enhancement for sharp, tight barriers.

6. **For which reaction is the Arrhenius plot more strongly curved at low temperature?**
   The Arrhenius plot for the $HNC \rightarrow HCN$ reaction is much more strongly curved at low temperatures, as shown in Figure 5.

8 Project 8: Minimum Energy Path estimation for CH_4 dissociation

The objective of this project is to model the homolytic cleavage of a single carbon-hydrogen bond in methane ($CH_4 \rightarrow CH_3^{\bullet} + H^{\bullet}$). The input code is written in `MOLPRO`. Below is a review of the critical sections of the provided input script:

```
memory,500,m
symmetry,nosym
basis=cc-pVDZ

geometry={
  angstrom
  C
  H1, C, rch
  H2, C, rch_eq, H1, ang
  H3, C, rch_eq, H1, ang, H2, 120.0
  H4, C, rch_eq, H1, ang, H2, -120.0
}

do rch_val = 0.1, 5.0, 0.1
  ...
  hf
  ...
  {multi
    closed,4
    occ,6
    wf,10,1,0
  }
  {rs2
    shift,0.2
  }
  CASPT2_ARR(i) = energy
enddo
```

The target C-H bond length is designated as the variable `rch`. The remaining bonds and angles are frozen at their equilibrium length (`rch_eq` = 1.09 Å).

A loop iterates the target bond length from 0.3 Å to 5.0 Å, completely mapping the Minimum Energy Path. At each step, three successive calculations are performed:

1. **Hartree-Fock (HF):** Generates the initial guess wavefunction.
2. **CASSCF (`multi`):** The Complete Active Space Self-Consistent Field method.
3. **CASPT2 (`rs2`):** Second-order perturbation theory is applied on top of the CASSCF wavefunction to recover dynamic electron correlation.

The energetic profile resulting from the scan loop is mapped in Figure 7.

The Potential Energy Surface effectively captures the complete thermodynamic topology of the bond dissociation process:

- **Repulsive Wall ($r < 1.0$ Å):** At extremely short internuclear distances, the energy sharply spikes due to massive internuclear repulsion and Pauli exclusion forces between the electron clouds.
- **Equilibrium Minimum ($r \approx 1.09$ Å):** The global energy minimum clearly aligns with the documented equilibrium C-H bond length of methane.
- **Dissociation Asymptote ($r > 2.5$ Å):** As the bond distance increases, the energy curve smoothly levels out into a horizontal plateau. This signifies successful homolytic cleavage into distinct $CH_3^{\bullet}$ and $H^{\bullet}$ radical fragments.

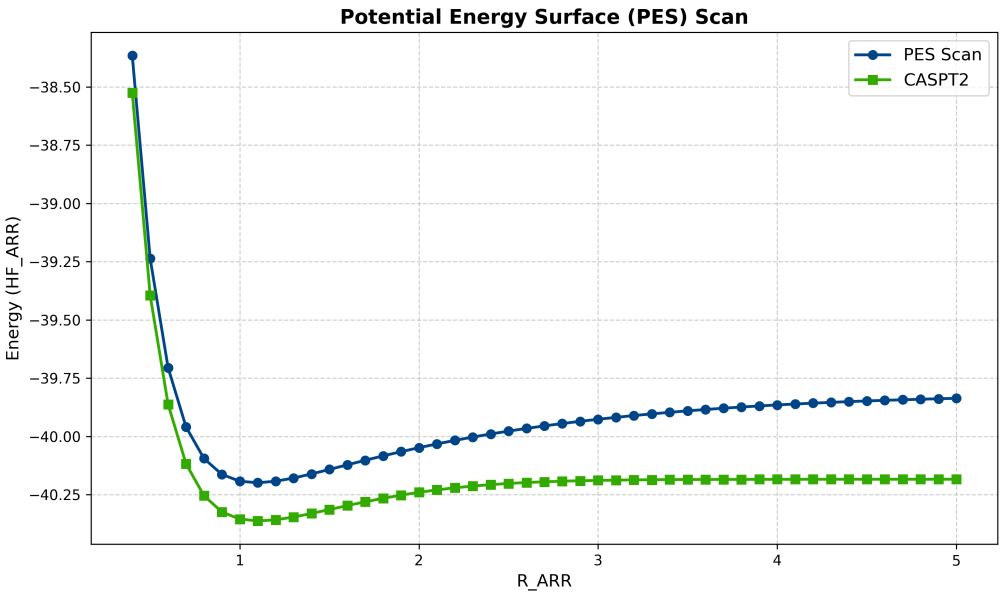


Figure 7: Potential Energy Surface (PES) scan mapping the dissociation of a C-H bond in methane. The x-axis tracks the C-H interatomic distance (Å), while the y-axis records the corresponding electronic energy.

The graph in Figure 7, shows that the CASSCF/CASPT2 framework successfully yields the physically correct asymptotic limit, demonstrating the absolute necessity of multi-reference methods for modeling chemical bond breaking, unlike standard HF calculations which often fail to represent correctly the energy pathway.

9 Project 9: Vibrational Spectroscopy

I performed the spectroscopy on a $(H_2O)_{11} + C_2H_6SO$ cluster using the guide presents at dragonball-vispec-orca setup.

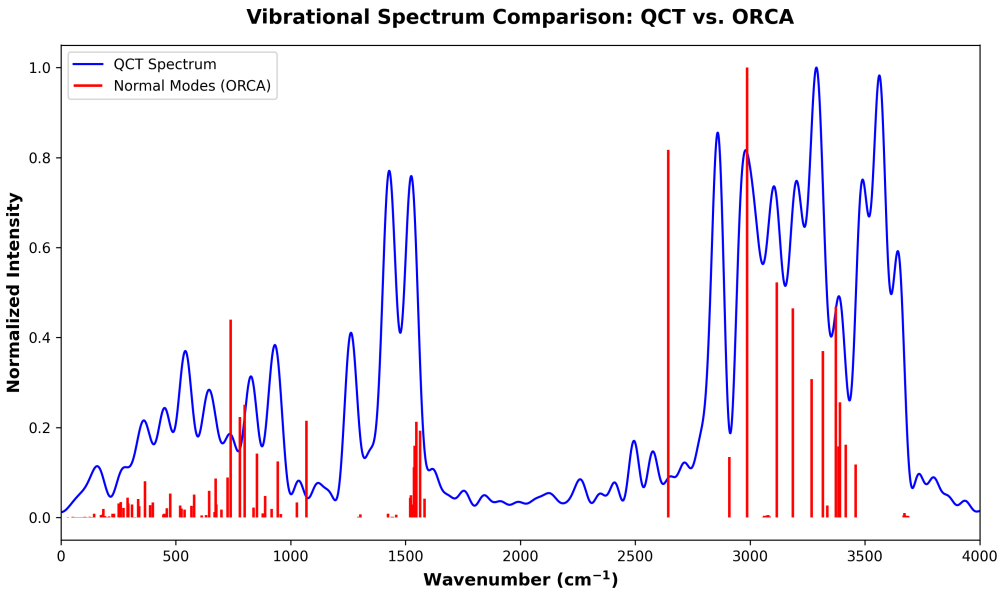


Figure 8: Comparison of the static harmonic spectrum (stick modes) and the continuous dynamic anharmonic spectrum (QCT MD) for the analyzed cluster.

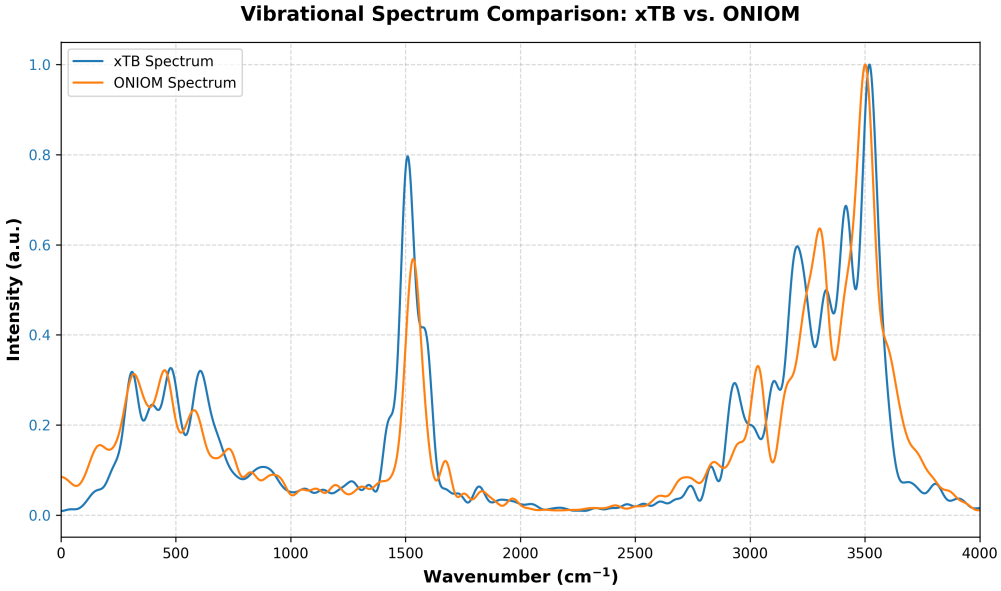


Figure 9: Comparison of the xTB and the *ab-initio* simulation.

Discussion Questions

1. **Harmonic vs. Anharmonic**
   There is a distinct **red shift** (a shift toward lower wavenumbers) when comparing the dynamic MD spectrum to the static harmonic xTB modes as shown in Figure 8. The harmonic approximation models the molecular bond as a perfect spring, which overestimates the restoring force at higher energy levels. In reality, chemical bonds are anharmonic; as the O–H bond stretches, the potential

energy well broadens and flattens, leading to lower vibrational frequencies. The molecular dynamics (MD) simulation inherently captures this anharmonicity resulting in significant peak broadening.

2. **xTB Dynamic Spectrum vs. High-Level Ab Initio Reference:**
   The xTB dynamic spectrum successfully reproduces the qualitative features of the high-level reference, capturing the distinct band regions and the general trend of the cluster environment as shown in Figure 9. Because xTB is a semi-empirical method, it may slightly shift exact peak positions or alter relative intensities compared to rigorous *ab-initio* methods.

   The **trade-off** here is computational efficiency versus absolute quantitative precision. Running ab initio molecular dynamics is computationally prohibitive for large clusters. xTB provides a remarkably accurate qualitative picture at a fraction of the computational cost, making extensive dynamic sampling feasible.

3. **Main Band Regions and Hydrogen Bonding:**
   The three main regions of the water cluster spectrum are:

   - **Librational modes (hindered rotations):** Broad band typically found $< 1000\,\mathrm{cm}^{-1}$, here the molecule is "trapped" by the hydrogen bonds of its neighbors.
   - **H–O–H Bending:** Narrower band located around $\approx 1600\,\mathrm{cm}^{-1}$, here the two hydrogen atoms swing toward and away from each other, while the oxygen atom stays mostly still.
   - **O–H Stretching:** Broad, intense region spanning $\approx 3200- \ 3600\,\mathrm{cm}^{-1}$, here the hydrogen atoms are actively pulled away from the oxygen atom and then slam back toward it, directly along the axis of the chemical bond..

   The O–H stretching modes and the librational modes are highly sensitive to the dynamic breaking and forming of hydrogen bonds. This causes a massive red-shift and broadening in the frequency. Simultaneously, the hydrogen bond network structurally restricts the free rotation of the water molecules, directly generating the hindered librational bands at lower frequencies.